



Nicolas Erbeti

- Robotics Engineer position -

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🌐 In a nutshell

- France, Japan & Luxembourg
- Worked in international and multilingual teams for ~4yrs.

💻 Computer Science

C/C++
★★★★★
Python
★★★★★
Javascript
★★★★★
C#
★★★★★
Java
★★★★★

🗣 Languages

French
★★★★★
Native language
English
★★★★★
Professional level
TOEFL score: 109/120
TOEIC score: 985/990
Japanese
★★★★★
Intermediate level
Equivalent JLPT N4
German
★★★★★
Certified level: European B1

Professional experiences

October 2023 up until now	Lead Autonomous Stack, Software Architect, Vision & Control Solarcleano Grass, Luxembourg Lead Autonomous Stack Engineer for TALOS, a Horizon Europe funded project Solarcleano / Energias de Portugal Grass, Luxembourg / Alqueva, Portugal • Leading a team of 3 person for the automation part of the work package 3.4 of the TALOS project, dedicated to cleaning a floating photovoltaic plant in Portugal. • Working with EDP and other members of the European Consortium, such as Alisys. • Wrote specifications and designed C++ code, state machine, and dual ROS1/ROS2 architecture; implemented kinematic models, UDP PLC communication, FRAMOS depth-cameras with YOLO+Apriltag perception pipeline, extended Kalman filter for localization, PID-based navigation, ROS websocket-based UI for video feedback and remote control, and an MQTT-enabled planification node. • Training a reinforcement learning policy with a sim2real approach for row switch.
July 2024 up until now	Project coordinator for the automation of a differential-drive robot Solarcleano / University of Luxembourg Grass / Luxembourg City, Luxembourg • Communicating with the team at the University of Luxembourg in charge of the software of a Jetson Orin-powered AGV following autonomously another robot. • Sharing our codebase for differential-drive robots to jumpstart the development. • Supplying them when they need specific hardware (plc, camera, lidars...)
April 2025 up until now	Computer vision-based control of a 1-ton autonomous robot Solarcleano Grass, Luxembourg • Established a solar panel segmentation pipeline in ROS1 and C++ with the PCL library and its RANSAC algorithm to process Lidars data. Filtered the raw results with Kalman filters, MonteCarlo estimations and geometrical constraints. • Developed state machine architectures and PID controllers with ROS actions. • Used the yaw of the segmented panel and a PID to write a realignment node for the navigation stack. Used the panel tilt with PIDs to regulate the robot's brush. • Created a mapping tool for photovoltaic plants with JavaScript and Maps API.
October 2023 to January 2025	Process improvements Solarcleano Grass, Luxembourg • Set up a docker image to develop ROS melodic code on Ubuntu 22.04 laptops. • Introduced rostest with our ROS packages to write C++ unit tests with GTest. • Set up a CI/CD pipeline with github actions to automatically run said tests.
October 2023 to March 2024	Reinforcement learning framework for the control of a robotic arm Omron Sinic X Tokyo, Japan • Contributed to a dual simulator framework and used it to teach a 6-dof Universal Robot 5 to autonomously cut through soft food with RL methods over a ROS network. • Calibrated this system with optimization tools (ADAM, Optuna...) to reproduce realistic force/torque data profile obtained with real-world sensor. • Developed a stable diffusion model to augment the training dataset of the Q-Policy. • Co-authored a paper on the topic which won the prize for best poster at ICRA 2024.
January 2023 to August 2023	Data fusion for the control of autonomous underwater & surface robots Notilo Plus Marseille, France • Developed automatic yaw, altitude, and position PID controllers in Python/C++ on ROS for the missions of an AUV; using GPS, pressure, and bathymetric sensors. • Conducted the unit and integration tests first in the Gazebo simulation and workbench before validating them in real conditions at Marseille's harbor. • Designed a state-space controller to remote-control an holonomic ASV.
April 2022 to August 2022	Integration of an ultrasonic sensors belt to a mobile robot Enova Robotics Ivry-sur-Seine, France • Conceived a sonar system as well as a safer mobility solution within ROS for the PGUARD robot. I also designed the mechanical/electrical parts of the prototype. • Integrated the Ouster OS0 Lidar both in the Gazebo simulation and in reality.
March 2021 to August 2021	

Tools

- ROS / ROS2 / Gazebo / RQT / Rviz / Foxglove
- Docker / Docker Compose
- Nvidia Jetson / Arduino / Raspberry Pi / MiTAC
- OpenCV / OpenPCL / Apriltag / YOLO
- OpenVino / Pytorch / Optuna / LabelStudio
- WSL / Ubuntu / Yocto
- Github / Gitlab
- VS Code / Gimp
- Solidworks / Matlab

Interests

Japanese language

I've been learning by myself during my spare time. My current goal is to pass the JLPT N3 exam.

Sports

I love hiking, cycling, kayaking... Any sort of group activity in natural landscapes. Also, I started taking Taekwondo classes last year.

Novels

I enjoy solving Agatha Christie's murder mysteries. Finally, I recently picked up on the books from "The Way of Kings" by Brian Sanderson.

Robot collection



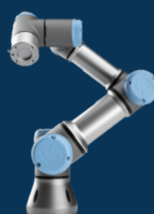
P-Guard



S-Drone



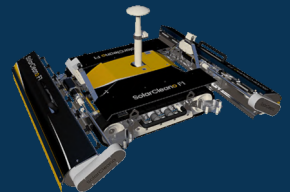
S-Buoy



Ur5e



B1A



F1A

Projects

From June 2025
up until now

Deployment of an online ROS laboratory

[Personal project](#) | [Arlon, Belgium](#)

I used the roslib javascript library to connect a web application to a Gazebo simulation running on my private server. The website is deployed on Cloudflare and the rosbridge websocket connection is established with Cloudflare tunneling.

- Real-time remote control and video feedback of the simulated turtlebot.
- Apriltag detection and YOLO segmentation of other robots.

December 2021

3D drawing tool with Virtual Reality controllers

[Personal project](#) | [Thionville, France](#)

I used an open-source medical robotics library to retrieve the pose of my Oculus Quest 2 controllers and draw their position in a 3D Python plot.

October 2021
to February 2022

Preliminary design of the NOIRE nanosatellite swarm

[LESIA Laboratory](#) | [Meudon, France](#)

I contributed to the specifications of the deployment conditions of a cubesat swarm orbiting around the moon for interferometric measurements of FIR signals. I also modeled the evolution of the swarm configuration in Python.

November 2020
to January 2021

Unicycle robot SLAM and path-finding algorithm

[Polytech Sorbonne](#) | [Paris, France](#)

I designed the motion control software of a mobile robot which I tested in Gazebo. With the help of SLAM ROS libraries and RRT path-finding algorithms, I managed to control the platform navigating in a room in fully autonomous mode.

October 2020 to
January 2021

Implementation of a mobile robot with embedded sensors

[Polytech Sorbonne](#) | [Paris, France](#)

I programmed the drivers and navigation algorithm of a mobile robot using a Raspberry pi, hall effect sensors and odometric principles. I mounted a serial arm on it with a camera, using a Haar-Cascade-powered AI to track human faces.

September 2019
to June 2020

Design of a pan-tilt robotic system

[Générations Robots](#) | [Paris, France](#)

I worked as part of a team to design an IP67 motorized system with two rotation axes, capable of accommodating sensors for our client who wanted to integrate it into a mobile robot. I was especially working on the electrical blueprint.

January 2019
to May 2019

Prototyping of a robotic arm

[Polytech Sorbonne](#) | [Paris, France](#)

Within a team of Polytech students, I built a 2DOF parallel robotic arm from the CAD on Solidworks to the control algorithm on an Arduino. I used an IR sensor to plan the motion of its effector along the curves of an irregular wall.

Education

From 2022 to 2023

Linguistic and traineeship program "Vulcanus in Japan"

[EU-Japan Centre for Industrial Cooperation](#) | [Tokyo, Japan](#)

- 4 months of intensive Japanese courses at the Naganuma School with fellow European participants followed by an 8-month internship in a Japanese company.

From 2021 to 2022

Master's degree "Astronomical and Space-based Systems Engineering"

[Paris Observatory, Sorbonne University](#) | [Paris, France](#)

- Space-based embedded systems engineering, sensors, optics, machine learning (pattern recognition, optimization...), astrophysics, orbitography, radiation.

From 2018 to 2021

Robotics Engineering Degree

[Polytech Sorbonne Engineering School](#) | [Paris, France](#)

- *3rd year*: mobile robotics (ROS, SLAM, Kalman filter, pathfinding...), real-time and object-oriented programming, test-driven development, control of high-inertia robotic arms, AI, machine learning (deep neural network, SVM...).
- *2nd year*: ethics, analogical and numerical automatics, robotics modelization (Denavit-Hartenberg, forward and reverse kinematics, state-space representation, trajectory planification...), image and signal processing and computer vision.
- *1st year*: control and power electronics, mechanical conception, mathematics.